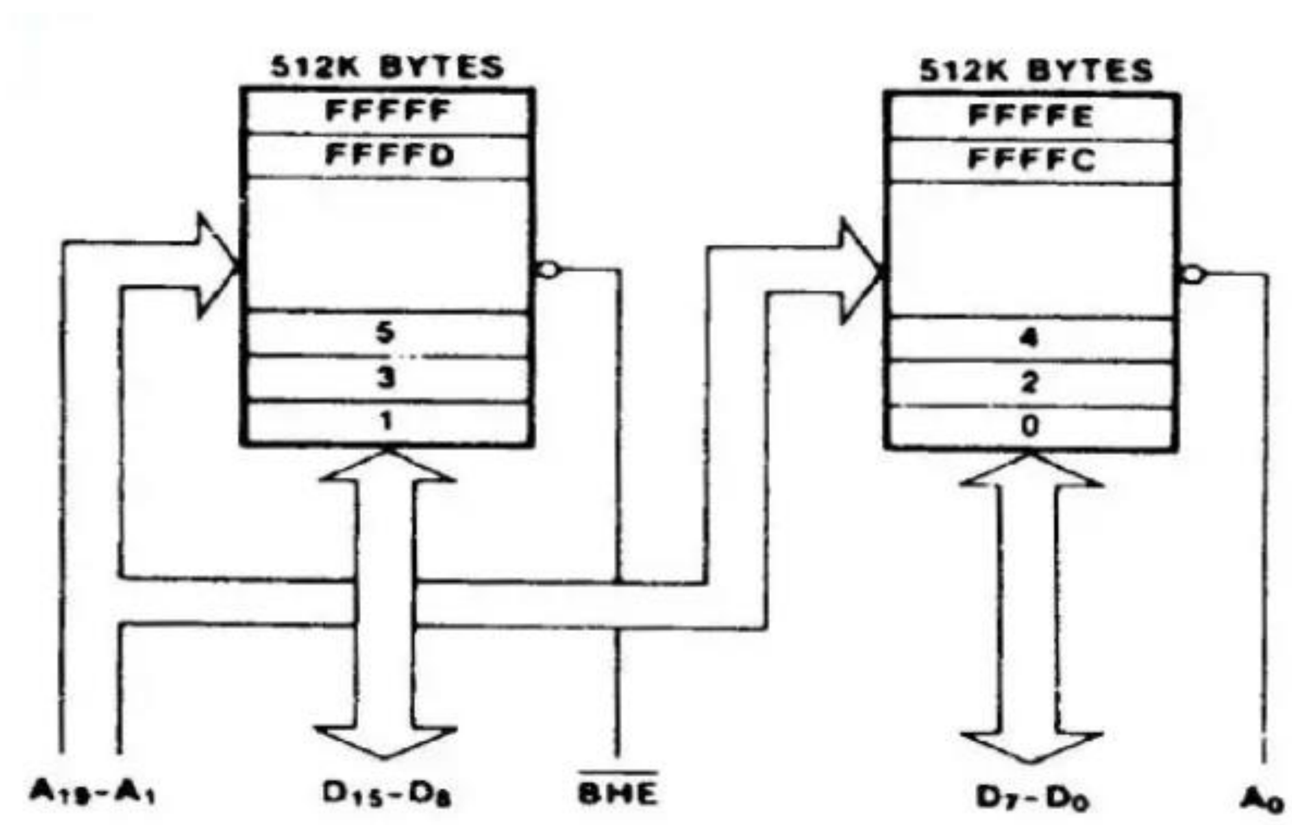
Memory banking in 8086

Concept of memory banking in 8086:

- Each memory location can store only one byte of data.
- But the data bus of 8086 processor has 16 data lines. In-order to utilize the data bus optimally the processor should access two consecutive memory locations ie 2 bytes of data in one cycle.
- However, if both these memory locations are in the same memory chip then they cannot be accessed at the same time. This is because the address bus cannot carry two addresses simultaneously.
- Thus the 8086 processor memory is divided into two banks where in each bank provides 8-bit data. (Note: banking concept is not there in 8085 and 8088 processor as their external data bus is 8 bits only)
- This concept is implemented by placing two consecutive locations on two memory chips.
- Physically, memory is implemented as two independent 512Kbyte banks: the low (even) bank and the high (odd) bank.

- Data bytes associated with an even address (00000H-FFFFEH) reside in the low bank, and those with odd addresses (000001H-FFFFFH) reside in the high bank.

- Address bits A1 through A19 select the storage location that is to be accessed. These are applied to both banks in parallel. **AO and bank high enable (BHE) are used as bank- select signals**.



- Each of the memory banks provides half of the 8086's 16-bit data bus. The lower bank transfers bytes of data over data lines DO through D7, while data transfers for a high bank use D8 through D15.

Table 1: Selection of odd and even memory bank.

$\overline{\text{BHE}}$	A0	Function
0	0	Choose both odd and even memory bank
0	1	Choose only odd memory bank
1	0	Choose only even memory bank
1	1	None is chosen

Question:

- How can 8086 read 8-bit data from an even address?
- How can 8086 read 8-bit data from an odd address?
- How can 8086 read 16-bit data from both even and odd addresses?

Explain with 8086 instructions.

<u>Read 8-bit data from even address memory bank</u>:

1. At first select, $\overline{BHE}$ **=1** and $\mathbf{A_0}$ **=0** in memory bank (shown in Table 1). Now even address bank of 8086 will be selected.
2. Find out the physical address (PA) of a particular memory location of even address memory bank.
3. Finally, a 8-bit data (2-bit Hex data) will be moved from that memory location to lower order byte of general purpose registers (AL/BL/CL/DL)

8086 instruction:
MOV AL, START[BX]
Suppose, [DS]=2000H, [START]=04H and [BX]=1200H
So, **PA = [DS]x10H+ ([START] + [BX]) = 21204H, which is an even address.**

Now consider, [21204H]=A9H. So that A9H data will be moved from the even address 21204H to AL. This is a read operation.

Read 8-bit data from odd address memory bank:

1. At first select, $\overline{BHE}$ **=0** and **A_0 =1** in memory bank (shown in Table 1). Now odd address bank of 8086 will be selected.
2. Find out the physical address (PA) of a particular memory location of odd address memory bank.
3. Finally, a 8-bit data (2-bit Hex data) will be moved from that memory location to higher order byte of general purpose registers (AH/BHL/CH/DH)

   8086 instruction:
   MOV AH, ARRAY[SI]

Suppose, [DS]=3000H, [ARRAY]=05H and [SI]=0300H
So, **PA = [DS]x10H+ ([ARRAY] + [SI]) = 30305H, which is an odd address.**

Now consider, [30305H]=2BH. So that 2BH data will be moved from the odd address 30305H to AH. This is a read operation.

Read 16-bit data from both even and odd addresses memory bank:

1. At first select, $\overline{BHE}$ **=0** and $\mathbf{A_0}$ **=0** in memory bank (shown in Table 1). Now odd address bank of 8086 will be selected.
2. Find out the physical addresses (PA1 and PA2) of two particular memory locations of even address and odd address memory bank.
3. Finally, a two 8-bit data (two 2-bit Hex data) will be moved from that memory locations to lower order byte (AL/BL/CL/DL) and higher order byte of general-purpose registers (AH/BHL/CH/DH) respectively.

8086 instruction:
MOV CX, DISPLACEMENT[SI]
Suppose, [DS]=5000H, [DISPLACEMENT]=06H and [SI]=0600H
So, **PA = [DS]x10H+ ([DISPLACEMENT] + [SI]) = 50606H,**
PA1= 50606H, which is an even address.
PA2= (50606+1)H= 50607H, which is an odd address.

Now consider, [50606H]=8CH and [50607]=2DH. So that 8CH data will be moved from the even address 50606H to CL and 2DH data will be moved from the odd address 50607H to CH. This is a 16-bit read operation.

Solve the following Home Work (HW):

How can 8086 write 8-bit data from an even address or 8-bit data from an odd address or 16-bit data from both even and odd addresses? Explain with 8086 instructions.